

Latent-Space Diagnostics

Two-Room’s encoder and predictor geometry: distances, dimensionality, and where τ_{fwd} ’s calibration bug actually comes from

Companion to `predictor-stitching.tex`

6 September 2026

1 Purpose

Before trusting any threshold built on top of the encoder or the predictor, it helps to look directly at the geometry they produce. This document collects diagnostic figures on Two-Room’s 27,675-landmark B0 landmark set, generated by `scripts/latent_diagnostics.py`. It is a companion to `predictor-stitching.tex`, which develops the IDM-ensemble/predictor verification mechanism these figures inform — in particular Figure 3 is the direct source of the τ_{fwd} mis-scaling diagnosed in that document’s E2 section, and Figure 8 is a visual confirmation of that document’s finding that ensemble variance does not discriminate real transitions from the wide candidate pool.

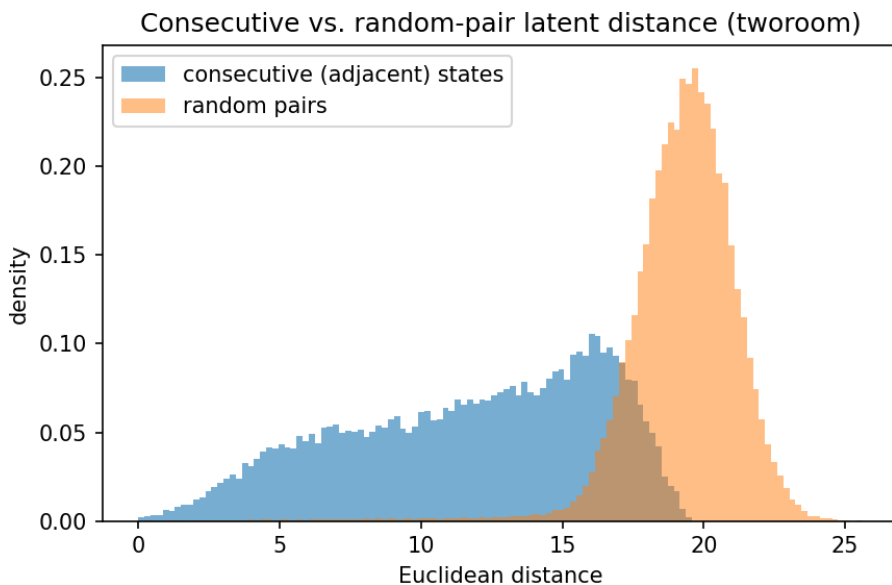


Figure 1: Consecutive (real, same-episode) states sit much closer together (mean 11.75, median 12.43) than random pairs (mean 19.25) — the encoder clearly separates temporal neighbors from arbitrary states, which is the entire precondition for a distance-based identification-edge threshold to mean anything.

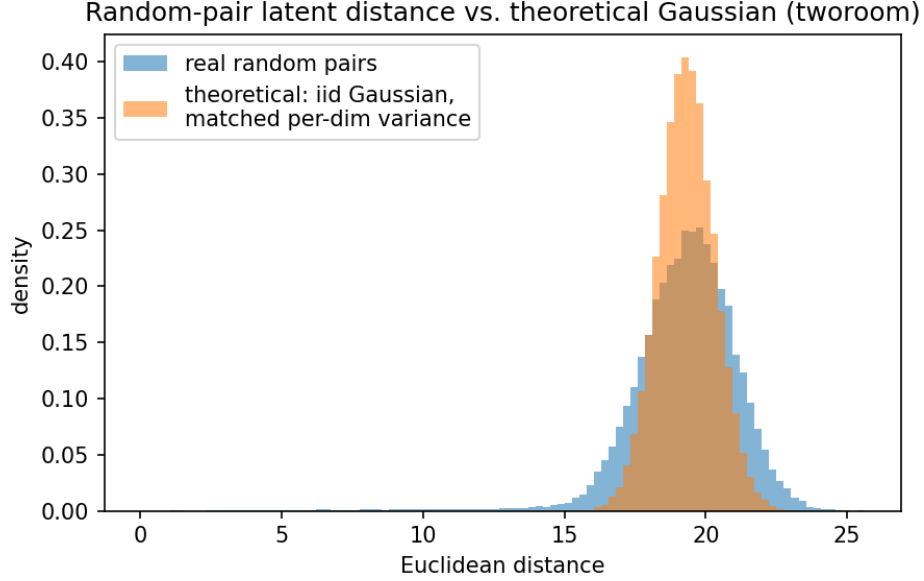


Figure 2: Real random-pair distances (mean 19.25) closely track a theoretical reference — two independent Gaussian vectors with the same per-dimension variance as the real embedding, sampled and measured the same way (mean 19.32). At the level of pairwise distance statistics, Two-Room’s embedding looks close to an anisotropic Gaussian blob, not a sharply-folded manifold — a sanity check behind `main.tex`’s calibration formula, which implicitly assumes something like this.

Correction (6 September 2026): the two figures below were originally built by feeding the predictor a single real action, zero-padded into the 10-dim slot `action_encoder` expects, and comparing its output one real step later. That invocation was wrong. The checkpoint’s actual training convention (`docs/lewm.txt`, Appendix D) applies a **frame-skip of 5**: one predictor call consumes the concatenation of **5 consecutive real actions** ($5 \times 2 = 10$ dims, exactly matching `action_encoder`’s input size with no padding at all) and predicts the state **5 real steps** later, not 1. Every figure below is now built the corrected way; see `scripts/latent_diagnostics.py`’s `build_block_history_chains`. The original padding-based numbers are kept in memory (`[[lewm-action-encoder-padding]]`) for the record, not repeated here.

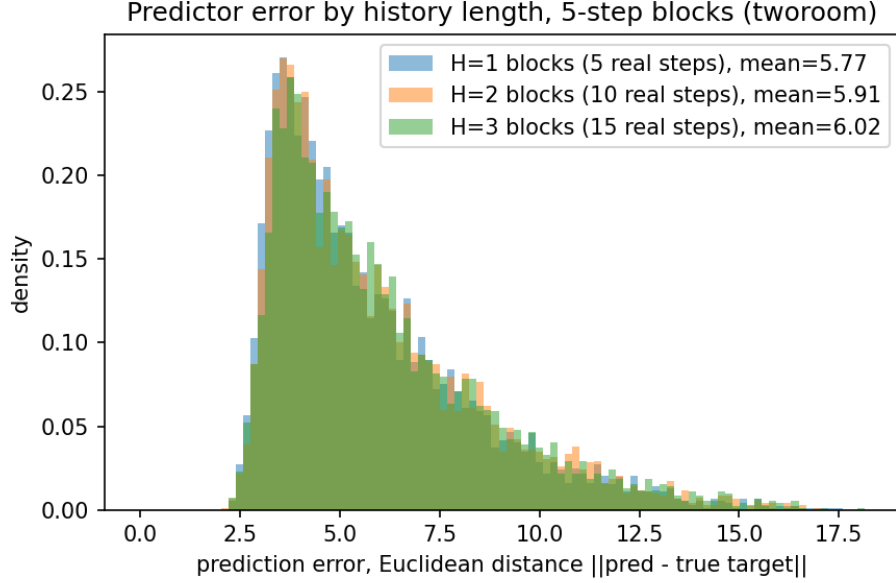


Figure 3: Predictor error (Euclidean distance between the prediction and the real target) using the CORRECT 5-action-block convention, at $H=1, 2, 3$ blocks of history (5, 10, 15 real steps of context respectively), on 5000 real chains per length. Error is essentially flat across H — means 5.77, 5.91, 6.02 — confirming (now on solid ground) that extra history does not sharpen the one-block prediction for Two-Room, consistent with its documented history length of 1 and its genuinely memoryless dynamics (position and action fully determine the next position).

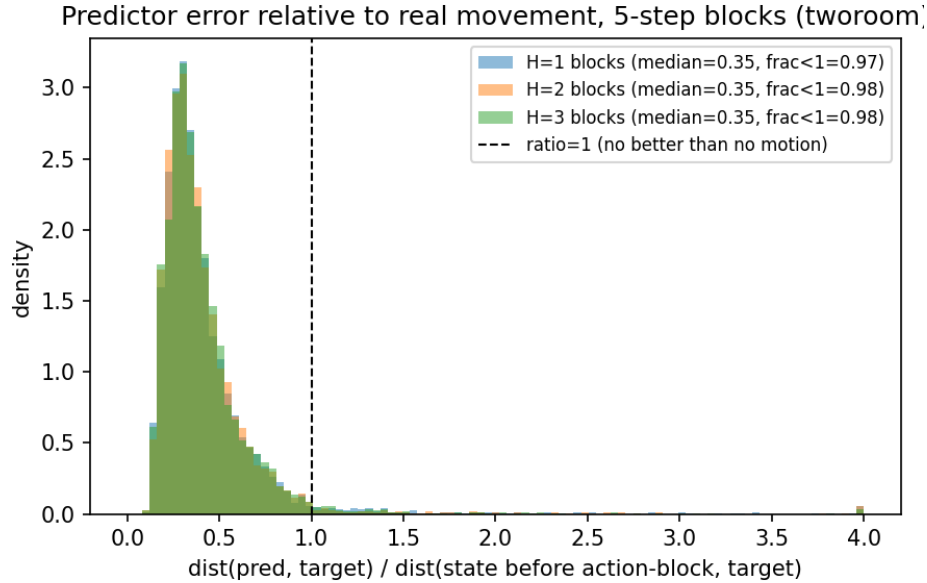


Figure 4: The same comparison, normalized: $\text{dist}(\text{pred}, \text{target}) / \text{dist}(\text{state before the block}, \text{target})$. Median ratio ≈ 0.35 at every H , with 97–98% of blocks showing *some* improvement and $\approx 78\%$ showing a strong one (< 0.5) — a real, substantial predictive signal, a sharp contrast with the flawed single-action version this replaces (which showed no signal at all under the wrong invocation).

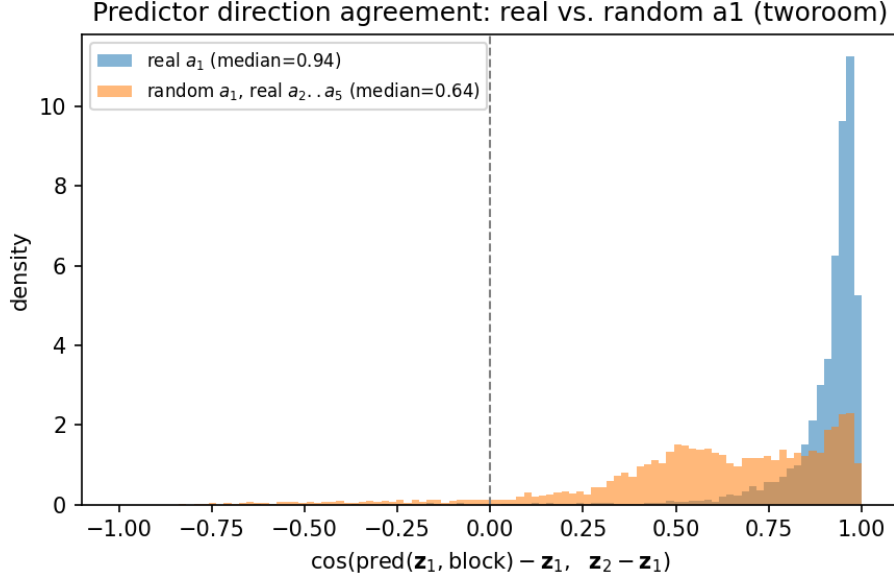


Figure 5: Direction agreement on a real one-block ($H=1$, 5-real-step) transition, with a control: $\cos(\text{pred}(\mathbf{z}_1, \text{block}) - \mathbf{z}_1, \mathbf{z}_2 - \mathbf{z}_1)$ using the real a_1 (mean 0.905, median 0.943, 98.4% above 0.5) versus replacing a_1 with a *random* real action while keeping $a_2, \dots, a_5$ exactly as they really occurred (mean 0.603, median 0.641, only 69.6% above 0.5). The real- a_1 numbers are even better than the earlier (wrongly-invoked) figure this replaces. More importantly, the drop under the random- a_1 control (0.943 $\rightarrow$ 0.641) confirms the model genuinely weighs the first action, not just the last four — if it didn’t, random and real a_1 would score about the same. This is the control that makes Figure 6’s positive result meaningful: the IDM’s inferred action being indistinguishable from a real one is not explainable by “any a_1 would look fine here.”

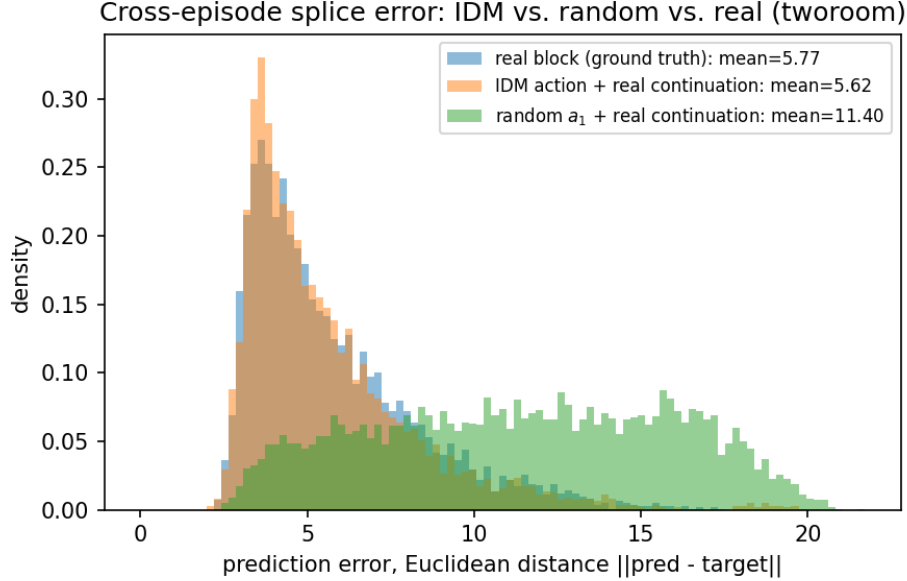


Figure 6: The real test this whole correction was for: does an IDM-inferred connecting action, for a candidate cross-episode pair $(\mathbf{z}_i, \mathbf{z}_j)$ from the wide top-20 pool, work as well as a genuine action once actually run through the predictor — and does it do *better* than an uninformed guess? Three conditions, all predicting from $\mathbf{z}_i$ against $\mathbf{z}_{j+4}$ (where j ’s own real rollout actually goes after its own next 4 actions): the hybrid block $(a, a_j, a_{j+1}, a_{j+2}, a_{j+3})$ with the IDM’s inferred a ; the same block with a replaced by a *random* real action (the control); and the reference (ground truth), Figure 3’s $H=1$ real block error. Result: **IDM hybrid mean 5.62, median 4.84 — indistinguishable from the real reference (mean 5.77, median 5.03) — while the random- a_1 control is mean 11.40, median 11.51, roughly double both.** This is the decisive version of the earlier figure: the IDM’s inferred action is not merely “fine because j ’s own continuation carries the block regardless of a_1 ” (Figure 5 already suggested this, but this figure confirms it directly in the actual cross-episode setting) — a wrong first action here is clearly and substantially worse, so the IDM’s specific inference is doing real, identifiable work.

Implication for predictor-stitching.tex’s E2 investigation. Every predictor call in that document’s stages 1 and 3 (the absolute floor, direction agreement, and relative-improvement criteria) used the same wrong single-action invocation this section just corrected. Those results are not automatically wrong — Two-Room’s numbers above barely changed under the fix, since its simple dynamics tolerate the malformed input reasonably well — but they were never validated under the correct convention either, and should be re-run with `build_block_history_chains`-style calls before being trusted as a final verdict on E2’s acceptance criterion.

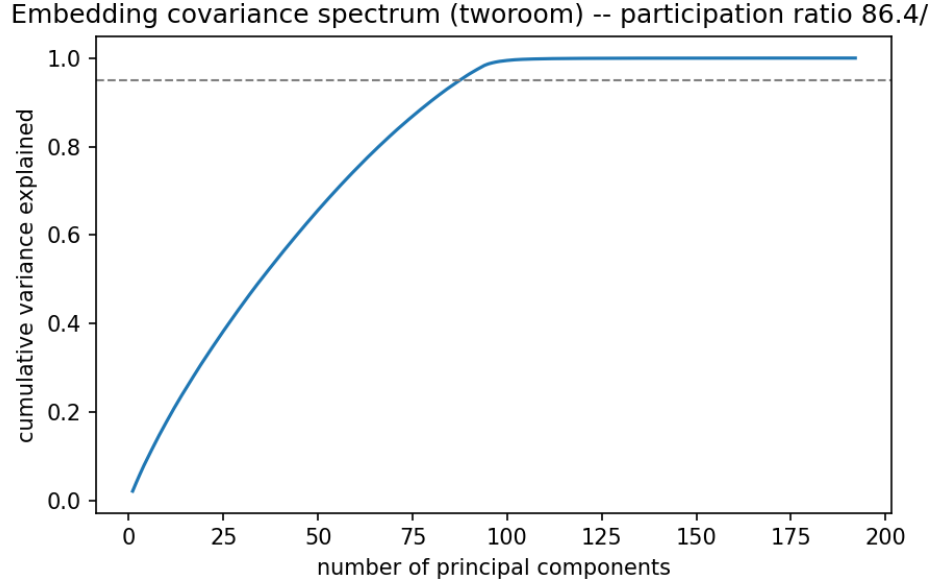


Figure 7: Cumulative variance explained by the embedding’s principal components. Participation ratio 86.4 of 192 ambient dimensions — comparable to what `main.tex` measured on Push-T (84 of 192), despite Two-Room’s calibration formula working and Push-T’s failing outright. Effective dimensionality alone does not predict which environment’s calibration holds; `main.tex`’s own diagnosis (its “Diagnosing why the calibrated threshold fails” section) pins Push-T’s failure on $\hat{\rho}$ itself, not on this.

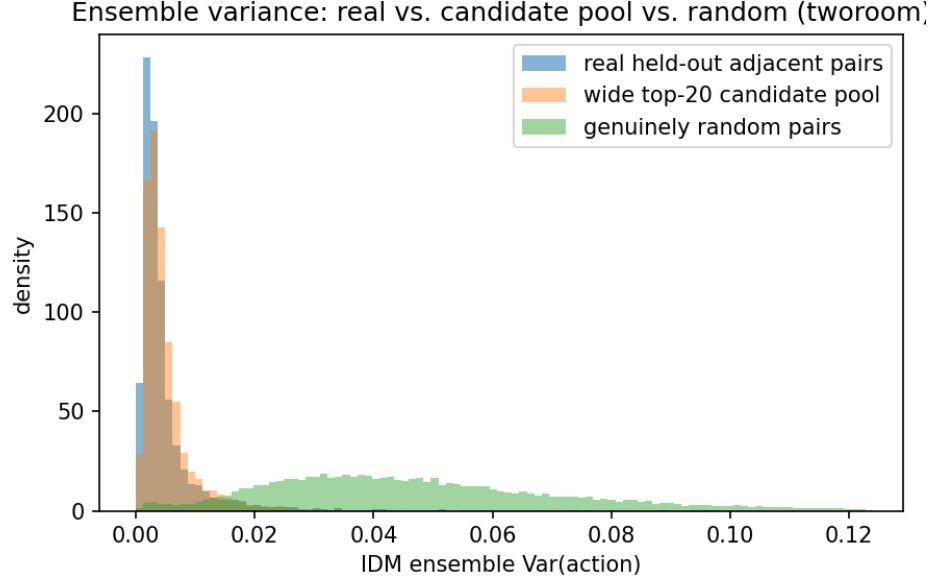


Figure 8: IDM ensemble variance on three populations: real held-out adjacent pairs (mean 0.00505, matching `predictor-stitching.tex`’s E1 result exactly), the wide top-20 candidate pool used in that document’s E2 section (mean 0.00560), and genuinely random, unrelated state pairs (mean 0.0495 — an order of magnitude higher). Real transitions and the candidate pool overlap almost completely, but both sit an order of magnitude below random pairs. This changes how E2’s finding should be read: variance is not an uninformative signal in general — it clearly separates “a coherent single action connects these states” from “these are just two unrelated states” — it simply has little left to discriminate *within* the top-20 pool, because that pool is already overwhelmingly made of genuinely connectable pairs, exactly as `predictor-stitching.tex`’s ground-truth environment check (its E2 stage 2) independently found.

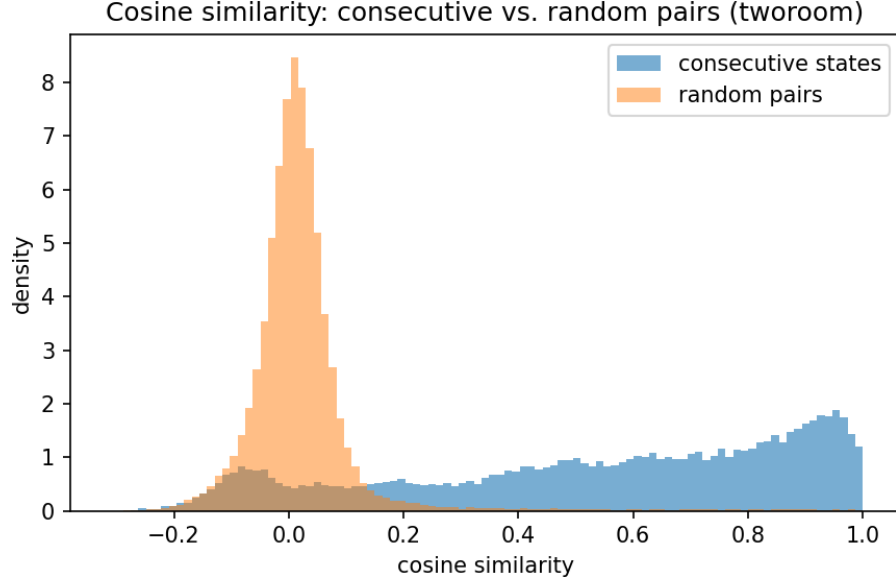


Figure 9: Cosine similarity between consecutive states (mean 0.540) versus random pairs (mean 0.021, near the zero expected for near-orthogonal random vectors in 192 dimensions). The adjacent-pair figure is close to $\hat{\rho}=0.584$ measured during calibration — consistent, since both summarize the same one-step correlation structure the calibration formula is built on.

Follow-up already run: Figure 5’s direction-agreement result was tried as a fix for `predictor-stitching.tex`’s E2 acceptance criterion (using the IDM’s inferred action on the candidate pool itself, gated on $\cos > 0.5$) and made precision worse, not better ($0.717 \rightarrow 0.609$) — a distance-bucket breakdown there shows direction is undefined for near-duplicate pairs (which dominate the pool) and non-target-specific at larger gaps, so Figure 5’s reliability on real transitions does not transfer directly to this candidate pool’s scale.

Other diagnostics worth having if this line of work continues: a version of Figure 5 computed per raw-distance bucket rather than pooled, to see at what scale (if any) direction agreement starts being genuinely target-specific rather than a generic bias; per-tier versions of Figures 1 and 8 once a real construction criterion is tested at scale (does the overlap in Figure 8 change with landmark count); the same figures repeated on Push-T, where Figure 2’s close match and Figure 7’s participation ratio are exactly the two things `main.tex` found behave differently; and a scatter of IDM ensemble variance against true oracle distance (rather than the marginal histograms here) to see directly whether any variance range corresponds to a reliable oracle-distance band, not just whether the two populations’ marginals overlap.